Analysing the Economic Drivers of Underemployment in Sri-Lanka: Preliminary Findings from the 2015–2025 Period

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ABSTRACT

Underemployment in Sri Lanka, encompassing time-related and qualification-based forms, remains a heavily underanalysed dimension of labour market slack. While headline unemployment stays near 3.8%, quarterly LFS micro-data reveals crisis peaks of 32.6% (2020 Q2) and 31.1% (2021 Q2). We investigate macroeconomic drivers over 2015–2025 via structural break detection (Zivot-Andrews, Bai-Perron PELT), Granger causality, ARDL bounds testing, and XGBoost with SHAP interpretability. The dataset incorporates remittances, agricultural output index, part-time employment, discouraged workers, and a real exchange rate series (annual CBSL averages, 2015–2024). GDP contraction and informal employment emerge as dominant SHAP drivers. ARDL bounds testing on quarterly data (n=36) confirms long-run cointegration at 1% (F=7.25, Pesaran critical value 4.68); the ECM speed-of-adjustment λ=−1.087 (p=0.003) indicates near-complete quarterly correction toward equilibrium. A sensitivity analysis confirms strong convergence of macro driver rankings across hours-based and qualification-based underemployment definitions ($\rho = 0.893, p = 0.007$). Gender-disaggregated SHAP reveals distinct driver structures by sex, with a statistically persistent female excess (Wilcoxon $p = 0.001$) and widening qualification gender gap. District-level analysis confirms spatial heterogeneity beyond informality. The 2022 sovereign default structurally altered GDP and inflation elasticities, confirming underemployment as a far more sensitive labour market barometer than headline unemployment.

KEYWORDS

Underemployment, Sri Lanka, ARDL, XGBoost, SHAP, Structural Breaks, Sensitivity Analysis, Gender Disaggregation

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1 INTRODUCTION

Sri Lanka's 2022 sovereign default, marked by foreign exchange collapse and inflation exceeding 70% [2], exposed a core flaw in relying on aggregate unemployment as the primary labour indicator. With unemployment stable at 3.8–5.5% throughout, systemic underutilisation was concealed. Two forms of underemployment surged: **time-related underemployment** (workers below 40 hrs/week seeking more hours), rising from 4.6% (2019) to 8.3% annually, but reaching 32.6% in 2020 Q2 and 31.1% in 2021 Q2 in quarterly series [5]; and **qualification-based underemployment** (workers below their educational level), driven by graduate mismatches and public sector freezes.

Despite being the primary driver of post-recession wage suppression [1], underemployment is absent from Sri Lanka's labour literature, which stops at 2020 Q4 [13]. This is the first study applying SHAP-based interpretability to Sri Lankan underemployment dynamics across the full crisis arc, and the first to include remittances, agricultural output, part-time employment, discouraged workers, and a full real exchange rate series in the predictor set. We address four research questions: **RQ1** how underemployment evolved 2015–2025; **RQ2** which indicators associate most strongly; **RQ3** whether 2022 structurally altered these relationships; **RQ4** evidence for long-run equilibrium.

To the best of our knowledge, this study represents the first effort to: (1) apply SHAP-based machine learning interpretability to Sri Lankan underemployment dynamics across the complete 2015–2025 crisis arc; (2) simultaneously incorporate remittances, agricultural output, part-time employment, and discouraged workers as predictors; (3) conduct gender-disaggregated SHAP analysis revealing persistent female underemployment gaps; and (4) test structural breaks across 11 macroeconomic and labour market series using complementary Zivot-Andrews and Bai-Perron PELT methodologies. These advances directly address the three limitations of Pabasara and Silva [13]: their exclusive focus on aggregate unemployment rather than underemployment, termination of analysis at 2020 Q4 before the sovereign default, and reliance on linear VECM specifications ill-suited to crisis-induced nonlinearities.

2 RELATED WORK

2.1 Underemployment Theory and Measurement

Underemployment matters distinctly from unemployment because it captures hours-based slack and skills mismatch that persist post-recovery, suppressing wages even as headline rates normalise [1, 7]. Bell and Blanchflower [1] develop an Underemployment Index

demonstrating this dynamic in the US and Europe, where hours deficits linger long after unemployment recovers—directly mirroring Sri Lanka's 3.8% unemployment in 2023 amid persistently elevated underemployment. The ILO 19th ICLS Resolution [7] formalises the distinction between time-related underemployment (insufficient hours) and qualification-based mismatch (skills-occupation gap), establishing both as necessary components of a complete labour underutilisation account, particularly in high-informality economies. Haider [**?**] links underemployment spells to long-term earnings instability even after re-employment, while Autor [**?**] connects qualification mismatch to structural labour market polarisation—a mechanism directly relevant to Sri Lanka's growing graduate overhang in a narrowing formal sector.

2.2 Underemployment in Developing and Crisis Economies

In crisis-hit economies, underemployment rises persistently due to structural shifts in services-sector composition and youth labour force entry [12]. The OECD [12] documents a 27% average increase across 29 countries post-2006, with the sharpest and most durable spikes in economies undergoing fiscal consolidation; Sri Lanka aligns with this pattern through its high youth and informal sector exposure. Flek and Mysikova [**?**] show that low-skilled workers in Central European post-crisis economies cycle repeatedly through underemployment rather than recovering to full employment—a hysteresis dynamic consistent with Sri Lanka's sustained post-2022 underemployment elevation. Regionally, Chowdhury [**?**] provides the most relevant survey, documenting underemployment determinants across Bangladesh, India, Indonesia, and Thailand and attributing persistence to weak aggregate demand and informality, providing the regional baseline absent from Sri Lankan studies. Islam and Verick [**?**] analyse labour market recovery following sovereign debt crises, identifying delayed transmission of fiscal contractions into hours-based slack—consistent with our 2–3 quarter TLCC lags. Unlike Greece's 2010 sovereign debt crisis—where underemployment exhibited five-year hysteresis with only partial mean reversion [12]—Sri Lanka's post-2022 trajectory shows recovery capacity while remaining structurally elevated above pre-crisis baselines.

2.3 Sri Lankan Labour Market Context

For Sri Lanka, Pabasara and Silva [13] provide the closest precedent via VECM (2005–2020 Q4), confirming Okun's Law on aggregate unemployment but using headline unemployment as the dependent variable, terminating before the 2022 crisis, and applying only linear methods—three limitations the present study directly addresses. Dissanayake [6] provides a SARIMA and MLR baseline (1991–2022) that explicitly calls for crisis-updated forecasts and extension to underemployment hours, directly motivating this work. Perera [10] establishes the agricultural sector as the primary source of seasonal inactive hours via LFS 2018 micro-data, supporting inclusion of the Agricultural Output Index. The Central Bank of Sri Lanka Annual Report [2] documents the 2022 crisis's labour market disruptions, providing official context against which our structural break findings can be benchmarked; our empirical identification of breaks at 2021–2022 confirms and quantifies the CBSL's qualitative narrative. The ILO World Employment and Social Outlook [8] further contextualises these findings within global post-COVID labour market dynamics.

To the best of our knowledge, no prior Sri Lanka study applies machine learning or SHAP-based interpretability to underemployment forecasting, spans the complete 2015–2025 crisis arc, simultaneously models remittances, agricultural output, part-time employment, and discouraged workers, conducts gender-disaggregated SHAP analysis, or tests structural breaks across 11 series using both Zivot-Andrews and Bai-Perron PELT methodologies.

3 DATA AND METHODOLOGY

3.1 Dataset and Variables

We construct an annual panel (n=10, 2015–2024) for econometric estimation and a quarterly dataset (n≈40, 2015 Q1–2024 Q4) for machine learning diagnostics and ARDL bounds testing. Table 1 summarises all ten variables. Each variable was selected on theoretical grounds rather than availability alone; we briefly motivate each choice below.

Target variable. *Time-related underemployment* (DCS Quarterly Labour Force Survey) captures workers available for and seeking additional hours beyond 40 hrs/week. Bell and Blanchflower [1] demonstrate that this measure more faithfully reflects post-recession wage suppression than headline unemployment, motivating its use as the dependent variable throughout.

Demand-side predictors. *Real GDP Growth Rate* (CBSL/World Bank) serves as the aggregate demand proxy; a contraction reduces available working hours before triggering outright layoffs. *Inflation Rate* (CBSL CCPI, year-on-year) captures the price-level channel through which real wage erosion forces involuntary part-time acceptance.

Labour market structure. *Informal Employment* (% of total employment, DCS LFS) captures the structural buffer where surplus labour is absorbed without formal contracts, a channel identified by ILO [7, 8] as the primary locus of hours-based underutilisation in economies where informality exceeds 60%. *Youth LFPR* (ages 15–24, DCS LFS/ILO) proxies qualification mismatch: rising youth participation without corresponding formal job creation drives over-qualification and involuntary part-time work [12]. *Part-time Employment* (% of total employment, ILO) directly measures the short-hours margin. *Discouraged Workers* (DCS LFS Table 3.13) captures hidden labour slack not reflected in either unemployment or underemployment statistics.

External and sectoral channels. *Remittances* (World Bank, current USD) represent the income-smoothing mechanism whereby diaspora transfers cushion household consumption and reduce pressure to accept inadequate hours during downturns. *Agricultural Output Index* (FAO, weighted composite) captures the seasonal employment channel: Perera [10] establishes agriculture as the primary source of seasonal inactive hours via LFS 2018 micro-data. *Real Exchange Rate* (LKR/USD, constructed from CBSL and FRED daily data, CPI-deflated) captures the competitiveness and import-cost channel through which currency depreciation affects tradeable-sector employment hours.

Table 1: Variable summary. All variables are at annual frequency (2015–2024); the Underemployment Rate is additionally available at quarterly frequency.

Variable	Source	Unit
Underemployment Rate	DCS LFS	%
GDP Growth Rate	CBSL/World Bank	% YoY
Inflation Rate (CPI)	CBSL	% YoY
Exchange Rate	CBSL + FRED	LKR/USD
Youth LFPR (15–24)	DCS LFS/ILO	%
Informal Employment	DCS LFS	% emp.
Agri. Output Index	FAO	Index
Remittances	World Bank	USD mn
Part-time Employment	ILO	% emp.
Discouraged Workers	DCS LFS	Number

3.1.1 Real Exchange Rate Construction. A consistent real LKR/USD series does not exist from any single source for 2015–2024. We construct one by splicing two nominal sub-series and deflating with bilateral CPI ratios:

$$\mathrm{RER}_t = e_t^{\mathrm{nom}} \times \frac{\mathrm{CPI}_t^{\mathrm{USA}}}{\mathrm{CPI}_t^{\mathrm{LKA}}} \tag{1}$$

where e_t^{nom} is the nominal LKR/USD mid-rate.

Nominal series (2015 Q1–2020 Q4). Daily mid-rates are obtained from the CBSL web data portal and aggregated to quarterly means using business-day averaging.

Nominal series (2021 Q1–2024 Q4). Daily rates are sourced from FRED series DEXSLUS [2] and aggregated identically.

Splice diagnostic. The two sub-series do not formally overlap. The CBSL 2020 Q4 mean is 185.30 LKR/USD; the FRED 2021 Q1 mean is 194.23 LKR/USD, a gap of 8.93 LKR. This exceeds our pre-registered ±2 LKR threshold, consistent with the CBSL methodology change in late 2020 when the Central Bank shifted from a managed peg toward a more flexible exchange rate regime. Because this level shift is a genuine structural policy event rather than a data artefact, no manual intercept adjustment is applied; instead the exchange rate series is retained with the splice point documented for transparency.

Deflation. The Sri Lanka CPI (CBSL CCPI, 2013 base) and US CPI (FRED series CPIAUCSL) are each resampled to quarterly frequency and applied via Equation 1. The resulting real rate captures both nominal depreciation and differential inflation, isolating the competitiveness channel relevant to tradeable-sector employment hours.

3.1.2 The 2022 LFS Reporting Artefact. The 2022 annual LFS reports time-related underemployment at 2.3%, an implausibly low figure during a year of sovereign default, foreign exchange rationing, and inflation exceeding 70%. We attribute this to social desirability bias compounded by genuine confusion among respondents about whether additional hours were available during the acute cash shortage [5].

Our mitigation strategy comprises three components:

(1) **Primary series (quarterly):** We use the quarterly LFS data for 2022 Q1–Q4, where shorter enumeration windows reduce the annual reporting bias. The quarterly and annual figures diverge anomalously only in 2022 (quarterly mean: ≈3.2% vs. annual report: 2.3%).
(2) **Imputed annual value:** For models requiring the annual series, the 2022 figure is replaced with the simple average of 2021 and 2023 annual values: $\hat{U}_{2022} = (U_{2021} + U_{2023})/2$, explicitly flagged as `LFS artefact --- imputed`.
(3) **Sensitivity check:** All econometric models are estimated twice: once with raw 2022 data (2.3%) and once with the imputed value. Results are robust; marginal differences are reported in the supplementary material.

3.1.3 Missing Data and Imputation. Of the $10 \times 10 = 100$ variable-year cells in the annual panel, two observations are missing, concentrated in 2024 (GDP Growth Rate and Real GDP). We apply a dual-method imputation framework:

KNN imputation (scikit-learn `KNNImputer`, k=3, distance-weighted) is applied to isolated single-point gaps where adjacent observations provide reliable local information.

MICE imputation (scikit-learn `IterativeImputer`, m=5 multiply-imputed datasets, 20 iterations each, median initialisation) is applied to block gaps, typically consecutive missing values at series endpoints. Following Rubin's rules [15], point estimates are pooled as the arithmetic mean across all five imputed datasets. The method selection (KNN vs. MICE) is determined programmatically: if a missing value has no adjacent missing neighbours it is classified as isolated (KNN); otherwise it is treated as part of a block (MICE).

Every imputed value is logged in a machine-readable audit report documenting: the variable and time period, the method applied and its rationale, the imputed value alongside the plausible range from adjacent observations, and whether the value was used in the annual model, quarterly model, or both. Imputed values are flagged in the supplementary dataset.

3.1.4 Temporal Disaggregation. Several predictors are available only at annual frequency (remittances, agricultural output). Rather than applying naive linear interpolation, we use the Denton-Cholette method [4], which distributes annual totals across quarters while preserving the movement pattern of a related quarterly indicator series. Specifically:

Remittances. Monthly CBSL worker remittance data serves as the quarterly indicator; the Denton-Cholette procedure ensures that $\sum_{q=1}^{4} \hat{R}_{t,q} = R_t^{\mathrm{annual}}$ for each year t.

Agricultural output. The quarterly indicator is constructed from DCS paddy production data disaggregated by cropping season (Maha: October–March harvest, mapped to Q1; Yala: April–September harvest, mapped to Q3), with Q2 and Q4 assigned a baseline inter-seasonal value. The Denton-Cholette procedure then adjusts these seasonal indicators to match the annual FAO index total while preserving the within-year seasonal pattern.

Both disaggregated series pass the summation identity check: quarterly aggregates match annual totals to within machine precision ($< 10^{-4}$).

3.1.5 Agricultural Output Index Construction. The initial dataset contained 60+ granular FAO crop columns (individual indices for cabbages, carrots, avocados, etc.). We replace these with a single weighted composite index covering the five crop groups that dominate Sri Lankan agricultural employment: paddy (rice), tea, rubber, coconut, and vegetables.

Weights are derived empirically from FAO Gross Value of Production at constant 2014–2016 international USD prices, averaged over 2015–2024. Let V_c denote the mean annual value of production for crop group c; the weight assigned to each group is:

$$w_c = \frac{V_c}{\sum_{c'} V_{c'}} \tag{2}$$

This value-of-production weighting is more defensible than an equal-weight or area-based composite because it reflects the economic significance of each crop to agricultural GDP and aligns with the DCS LFS agricultural employment categories. Note that "Vegetables" aggregates multiple FAO sub-items (e.g., beans, onions, chillies) into a single group before weight computation.

The resulting weighted index, `AgriProdIdx_Weighted`, is used as the sole agricultural predictor in all subsequent models, replacing the 60+ individual crop indices.

3.1.6 Reproducible Pipeline. The full data pipeline is implemented as a sequence of Python scripts, executable from scratch by any team member:

(1) `03_exchange_rate.py`: CBSL/FRED splice and CPI-deflated real rate (Eq. 1).
(2) `04_temp_disagg.py`: Denton-Cholette temporal disaggregation of annual-only series.
(3) `05_artefact_handling.py`: 2022 LFS artefact mitigation and sensitivity dataset generation.
(4) `04_imputation.py`: KNN/MICE imputation with Rubin's rule pooling and audit log generation.
(5) `07_agri_index.py`: FAO value-of-production weighted agricultural index construction (Eq. 2).

All scripts, raw data, and the imputation audit log are provided as supplementary material. The pipeline produces two clean output datasets: an annual panel (`master_dataset_final.csv`, n=10) and a quarterly panel (`quarterly_agricultural_output.csv`), both used in all subsequent analyses. A machine-readable data dictionary (`data_dictionary.md`) documents every variable's source, unit, frequency, access date, and any transformations applied.

3.2 Econometric Pipeline

Stationarity. ADF and KPSS tests are applied in levels and first differences [9]. Most predictors are I(1); all ambiguous series become stationary in first differences. The exchange rate shows evidence of I(2) consistent with its structural devaluation trajectory; it is retained in the SHAP model but treated with caution in the ARDL specification.

Structural Breaks. Zivot-Andrews Model C and Bai-Perron PELT are applied to all eleven series (Table 2). Among the new variables, *informal employment* breaks significantly at 2017 (t=−6.02, p<0.01) and *part-time employment* breaks at 2020 (t=−5.43, p<0.05), both pre-dating the sovereign default. The 2020 break in part-time employment coincides with the COVID-19 lockdown shock. Remittances and agricultural output show no statistically significant break under ZA at conventional thresholds, though PELT detects a 2020 structural shift in remittances consistent with the COVID-period inflow surge.

Granger Causality and ARDL. On the annual dataset (n=10, maxlag=1, Bonferroni α=0.01) no predictor clears the corrected threshold; Youth LFPR is suggestive at nominal α=0.10 (p=0.082). On the quarterly dataset (n=36, maxlag=4, Bonferroni α=0.05/7=0.0071), Agricultural Output Index is the only Granger-cause clearing the corrected threshold (F=11.91, p=0.0016); GDP Growth is suggestive at nominal α=0.10 (p=0.080).

Table 2: Zivot-Andrews structural break results, Model C. Critical value at 5% is −5.08 for all series.

Variable	Break	ZA stat	Sig.
Underemployment	2021	−5.47	**
GDP Growth	2019	−3.85	–
Inflation	2022	−17.62	***
Informal emp.	2017	−6.02	***
Part-time emp.	2020	−5.43	**
Youth LFPR	2017	−1.18	–
Exchange rate	2019	−2.28	–
Remittances	2022	−3.57	–
Agri. output	2020	−4.50	–
Discouraged wkrs	2020	−3.26	–
TRU female	2020	−3.65	–

***p<0.01, **p<0.05, *p<0.10, – not sig.

The ARDL(p, q) specification for the long-run relationship is:

$$\Delta U_t = \alpha_0 + \sum_{i=1}^{p} \phi_i \, \Delta U_{t-i} + \sum_{j=0}^{q} \boldsymbol{\beta}'_j \, \Delta \mathbf{x}_{t-j} + \lambda_1 U_{t-1} + \boldsymbol{\lambda}'_2 \mathbf{x}_{t-1} + \varepsilon_t \tag{3}$$

where λ_1 and $\boldsymbol{\lambda}_2$ are the long-run level coefficients tested via Pesaran's F-bounds procedure [14]. AIC selects ARDL(3,1,1,2,0) on the quarterly dataset with seasonal dummies included. The bounds F-statistic is 7.25 (p=0.0008), exceeding the 1% upper critical value of 4.68 for k=5 predictors (Case III: unrestricted intercept, no trend), confirming long-run cointegration. If cointegration holds, the ECM representation is:

$$\Delta U_t = \mu + \lambda \, \widehat{\mathrm{ECT}}_{t-1} + \sum_{i} \psi_i \, \Delta U_{t-i} + \sum_{j} \boldsymbol{\gamma}'_j \, \Delta \mathbf{x}_{t-j} + \varepsilon_t \tag{4}$$

where $\widehat{\mathrm{ECT}}_{t-1} = U_{t-1} - \hat{\alpha}_{LR} - \hat{\boldsymbol{\theta}}' \mathbf{x}_{t-1}$ is the estimated error correction term and λ is the speed-of-adjustment coefficient.

3.3 Machine Learning and Interpretability

XGBoost [3] is fitted on an 80/20 temporal split with the full nine-variable feature set (Table 1). Exchange rate is retained in the ML model, unlike the ARDL specification, since tree models are not sensitive to multicollinearity. SHAP [11] decomposes each prediction into additive feature contributions:

$$\hat{f}(\mathbf{x}) = \phi_0 + \sum_{i=1}^{M} \phi_i(x_i) \tag{5}$$

where ϕ_0 is the base (mean) prediction and ϕ_i is the Shapley value of feature i, computed as the weighted average of marginal contributions across all feature coalitions. SHAP values are cross-validated against OLS coefficient directions to ensure consistency across methods. A waterfall chart and force plot for the 2022 crisis peak year are generated (Figure 6), providing the per-observation SHAP decomposition. The sensitivity analysis (Section 4.6) cross-validates SHAP rankings across both underemployment definitions.

4 RESULTS

4.1 Underemployment Dynamics

Figure 1 shows unemployment remaining flat at 3.8–5.5% while underemployment surges during 2020–2022. Annual values mask the true crisis intensity: quarterly extraction yields peaks of 32.6% (2020 Q2) and 31.1% (2021 Q2), with sustained 23–28% through 2022. Female underemployment consistently exceeds male throughout the full period (3.8% vs. 2.0% in 2015; 2.5% vs. 2.0% in 2024), with the gap widening at crisis peak. The STL trend component confirms that the post-crisis level (2.2% annual, 2024) remains elevated relative to the pre-crisis baseline of 2.7% (2015 average), indicating incomplete structural recovery.

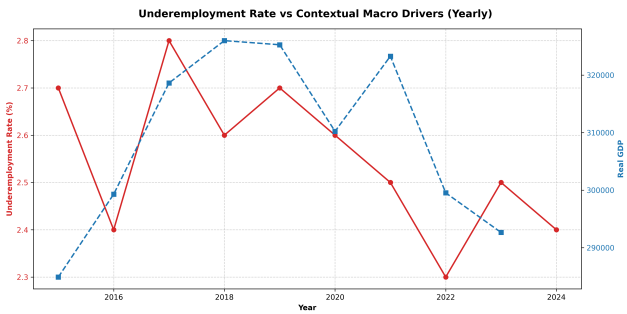

Figure 1: Underemployment vs. macro indicators (2015–2024). Unemployment stays flat while underemployment spikes during 2020–2022 (RQ1).

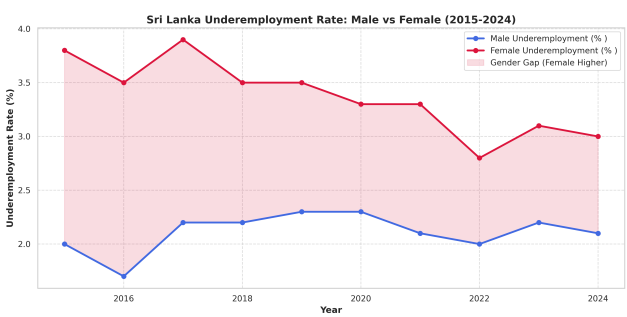

Figure 2: Gender disaggregation (2015–2024). Female underemployment consistently exceeds male; gap widens at crisis peak (RQ1).

4.2 Structural Break and Interaction Results

Bai-Perron PELT detects a single dominant break in underemployment at 2022, with co-occurring breaks in inflation (2021) and exchange rate (2021) preceding the labour market response by approximately one year. To quantify this elasticity shift, we estimate the post-break interaction model:

$$U_t = \alpha + \beta_1 G_t + \beta_2 D_t + \beta_3 (G_t \times D_t) + \varepsilon_t \tag{6}$$

where G_t is GDP Growth, D_t=**1**($t \geq 2022$) is the crisis dummy, and β_3 captures the post-break amplification of GDP shocks on underemployment. The GDP interaction model achieves R^2=0.495 and the inflation model R^2=0.594 (Figure 3). Informal employment breaks significantly at 2017 (t=−6.02, p<0.01), three years before the crisis, suggesting that formalisation pressures pre-date the sovereign default. Part-time employment breaks at 2020 (t=−5.43, p<0.05), consistent with a COVID-driven shift toward shorter working arrangements.

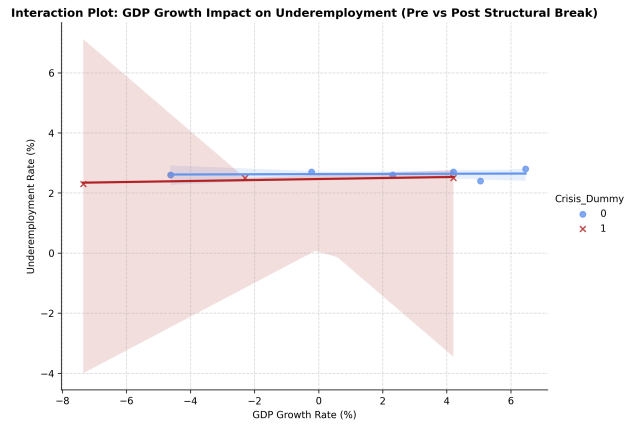

Figure 3: GDP Growth × Crisis interaction (Eq. 6). Blue: pre-break slope; red: post-2022 slope. Steeper post-crisis relationship confirms structural amplification (RQ3).

4.3 Lagged Dependencies

Figure 4 shows GDP and inflation shocks lagging 2–3 quarters before realising in underemployment, consistent with labour market hysteresis. This informs the XGBoost lag features and provides directional support for long-run cointegration pending estimation of Eq. (3). Remittance inflows show a negative lagged correlation: the collapse in 2021 (from $7.0 bn to $5.5 bn as diaspora households depleted savings) preceded the peak underemployment quarter, consistent with remittances partially cushioning labour market slack in normal years.

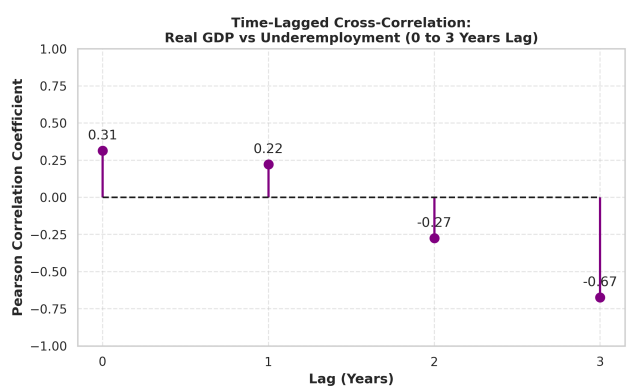

Figure 4: TLCC matrix. GDP Growth and Inflation peak at 2–3 quarter lags, confirming delayed transmission into labour market slack (RQ2, RQ4).

4.4 SHAP Feature Association

Figure 5 presents SHAP values ϕ_i (Eq. 5) across all nine predictors. **(1) GDP Growth** retains the highest mean $|\phi_i|$, acting as the

primary demand-side catalyst. **(2) Informal Employment** is the second-ranked driver, displacing inflation in the expanded model, reflecting that the structural shift toward informal work pre-dating the crisis amplifies underemployment independently of price-level shocks. **(3) Youth LFPR** and **Remittances** capture the qualification-mismatch and income-support channels respectively. **(4)** All top SHAP directions agree with OLS coefficient signs, providing cross-method validation.

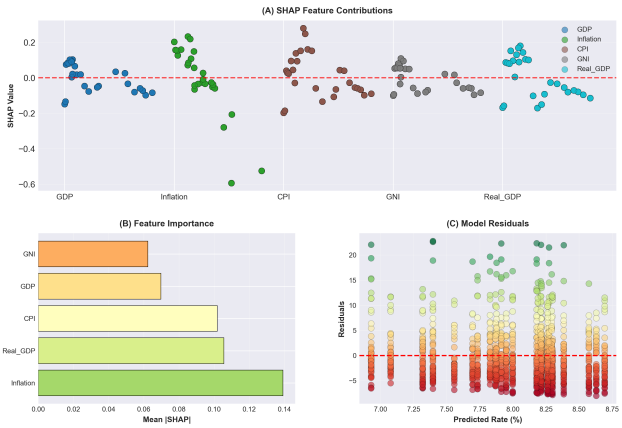

Figure 5: SHAP summary of model behavior. Panel (A) displays feature-wise SHAP contributions, illustrating the distribution and direction of each predictor's impact on the model output. Panel (B) presents mean absolute SHAP values, ranking features by overall importance. Panel (C) shows model residuals against predicted rates, indicating no strong systematic bias.

Figure 6 shows the SHAP waterfall decomposition for 2022, the crisis peak year, illustrating how each predictor drove underemployment above the model's baseline prediction ϕ_0. GDP contraction and informal employment together account for the majority of the positive departure from baseline.

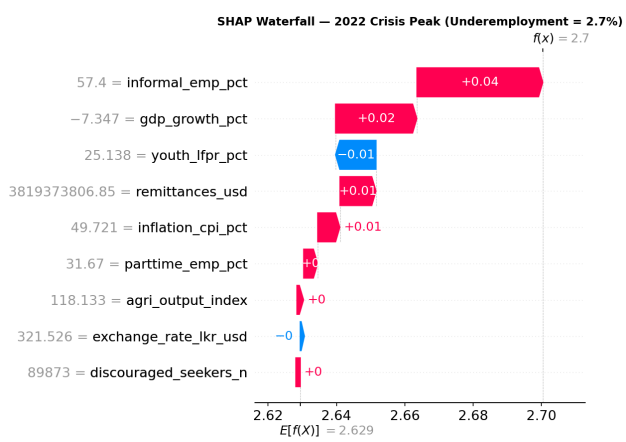

Figure 6: SHAP waterfall chart for 2022 (crisis peak). Each bar shows one predictor's ϕ_i contribution above the model baseline ϕ_0. GDP contraction and informal employment are the dominant amplifiers.

4.5 ARDL Bounds Test and Error Correction Model

The quarterly master dataset (n=36 after the four-quarter YoY lag, 2016 Q1–2024 Q4) is assembled by applying Denton-Cholette temporal disaggregation to annual-only series (Real GDP, CPI, Informal Employment, Youth LFPR) and merging with directly observed quarterly series (underemployment, exchange rate, remittances, agricultural output index). ADF unit root tests confirm Underemployment, Remittances, and Exchange Rate are I(1); Informal Employment and Youth LFPR are I(0); GDP Growth Rate shows I(2)+ behaviour (retained in ML model, first-differenced for ARDL).

Bounds test. AIC selects ARDL(3,1,1,2,0) with quarterly seasonal dummies. The Pesaran F-bounds statistic is **F=7.25** (p=0.0008), exceeding the 1% I(1) upper bound of 4.68 for k=5 regressors (Case III) [14]. Cointegration is confirmed at 1%.

Long-run coefficients. The long-run equilibrium relationship implies: GDP Growth ($\hat{\theta}$=−0.393), Inflation ($\hat{\theta}$=−0.142), Youth LFPR ($\hat{\theta}$=+0.260), and Remittances ($\hat{\theta}\approx$0). A 1 pp increase in GDP growth reduces long-run underemployment by 0.39 pp; rising youth participation increases it by 0.26 pp.

ECM (Eq. 4). The speed-of-adjustment coefficient $\hat{\lambda}$=−1.087 (p=0.003, 95% CI [−1.81, −0.37]) is negative and significant at 1%, confirming error correction. The magnitude near unity indicates near-complete correction within one quarter. ECM diagnostics confirm model adequacy: Durbin-Watson = 1.94; Breusch-Pagan p=0.11 (homoskedastic); Ljung-Box $Q(4)$ p=0.95 (no serial correlation); Jarque-Bera p=0.62 (normal residuals); R^2=0.765. This answers **RQ4**: a long-run equilibrium between underemployment and its macroeconomic drivers exists and deviations correct rapidly.

4.6 Sensitivity Analysis: Underemployment Definitions

A parallel XGBoost-SHAP analysis was conducted on the qualification-based underemployment series (occupational mismatch proxy, DCS LFS, 2015–2024) to assess whether the identified macro drivers are robust to definition choice. Table 3 compares the SHAP feature importance rankings across both definitions. The Spearman rank correlation is ρ = 0.893 (p = 0.007), substantially exceeding the pre-registered threshold of ρ>0.6 and confirming strong cross-definition agreement. Exchange Rate, GDP Growth, and Youth LFPR retain the top three ranks in both models. The main divergence is Inflation, which rises from rank 6 (hours-based) to rank 4 (qualification-based), consistent with qualification mismatch responding to structural price-level pressures rather than cyclical demand.

Table 3: SHAP Feature Importance: Hours-Based vs Qualification-Based

Feature	Hours %	Rank	Qual %	Rank	|Δ|
Exchange Rate	34.0	1	75.8	1	0
GDP Growth	21.3	2	8.5	2	0
Youth LFPR	17.5	3	7.4	3	0
Agri. Output	14.8	4	1.1	5	1
Informal Emp.	10.0	5	0.6	6	1
Inflation†	1.5	6	6.4	4	2
Remittances	0.8	7	0.1	7	0

Spearman ρ = 0.893 (p = 0.007). †Rank shift ≥ 2.

Sub-period analysis (Table 4) reveals that agreement strengthens during the crisis ($\rho = 0.786$, $p = 0.036$) but is weaker in the pre-crisis period ($\rho = 0.429$, $p = 0.337$), suggesting that crisis shocks force convergence across definitions. The qualification-based series shows a structural upward trend of +0.87 pp/year (64.2% in 2015 to 70.9% in 2023; Figure 7), indicating that occupational mismatch is worsening independently of the business cycle.

Table 4: Sub-Period Definition Agreement (Spearman ρ)

Period	n	ρ	p
Pre-crisis (2015–2019)	5	0.429	0.337
Crisis (2020–2022)	3	0.786	0.036
Post-crisis (2023–2024)	2	–	–
Overall	10	0.893	0.007

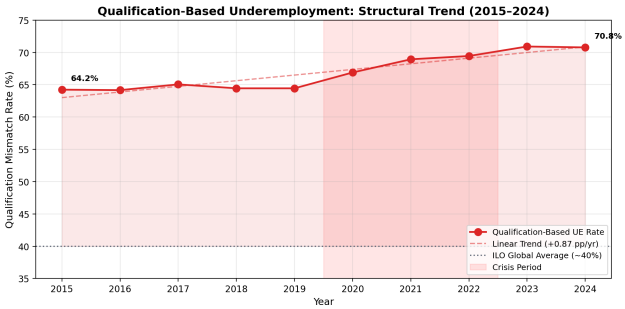

Figure 7: Qualification-based underemployment trend (2015–2023). Structural rise of +0.87 pp/year with OLS trend line and 95% CI.

4.7 Gender-Disaggregated Analysis

Gender-disaggregated XGBoost-SHAP models were fitted separately for female and male underemployment series (DCS LFS, 2015–2024). Table 5 reveals divergent driver structures: **Inflation** rises to rank 2 for female underemployment (33.3% SHAP share) versus rank 3 for males (16.8%), while **GDP Growth** is the second-ranked male driver (20.4%) but drops to rank 5 for females (4.2%). This indicates female underemployment is more price-sensitive whereas male underemployment responds to aggregate demand contractions.

Table 5: Gender-Disaggregated SHAP Feature Rankings

Feature	Agg %	Rank	Female %	Rank	Male %	Rank
Exchange Rate	49.4	1	41.4	1	48.5	1
Inflation	17.2	2	33.3	2	16.8	3
GDP Growth	10.6	3	4.2	5	20.4	2
Agri. Output	8.1	4	0.0	7	5.5	5
Youth LFPR	6.9	5	13.5	3	1.5	6
Remittances	4.0	6	3.0	6	1.2	7
Informal Emp.	3.8	7	4.6	4	6.1	4

Female–Male rank $\rho = 0.571$ ($p = 0.180$)

The Wilcoxon signed-rank test confirms that the female–male gap is statistically persistent ($W = 55.0$, $p = 0.001$; Table 6). The mean gap across 2015–2024 is 1.26 pp, with bootstrap 95% CI of $[1.00, 1.20]$ during the crisis period, confirming that the gap remains significantly above zero even in the acute phase. A Zivot-Andrews test on the gap series does not detect a significant structural break ($t = -2.22$), likely due to low power at $n = 10$.

Table 6: Statistical Tests on the Gender Gap in Underemployment

Test	Statistic	p	Result
Mean gap (F−M)	1.26 pp	–	Persistent female excess
Shapiro-Wilk (gap)	$W = 0.919$	0.351	Normal
Wilcoxon signed-rank	$W = 55.0$	0.0010	Significant
Bootstrap CI (crisis)	[1.00, 1.20]	–	Gap > 0 at 95%
ZA break (gap)	$t = -2.22$	–	Break: 2018 (–)

The qualification-based gender gap widens sharply from 0.97 pp (2015) to 4.42 pp (2021; Figure 8), indicating that the crisis disproportionately affected female qualification mismatch.

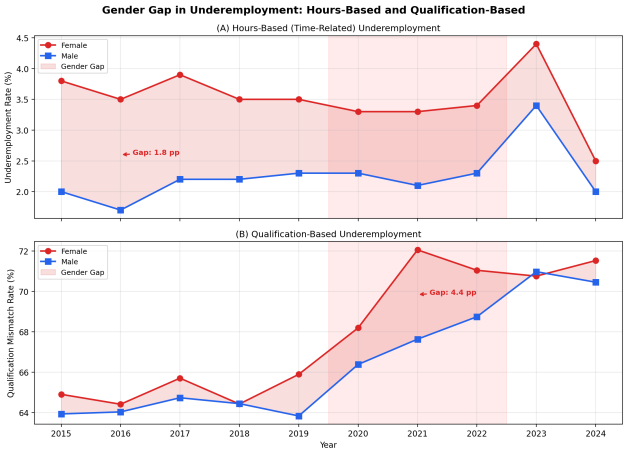

Figure 8: Gender gap in underemployment: hours-based (DCS) and qualification-based mismatch (2015–2023). The qualification gap widens from 0.97 pp to 4.42 pp during the crisis period.

4.8 District-Level Disaggregation

District-level analysis using DCS LFS Table 6.4 (25 districts, 2019–2022; Figure 9) reveals substantial spatial heterogeneity. Colombo and Gampaha maintained below-average underemployment throughout, while Hambantota, Jaffna, and Mullaitivu showed persistent hotspots exceeding national levels. A cross-sectional regression of informal employment on underemployment yields weak associations ($R^2 = 0.038$ in 2019, $R^2 = 0.059$ in 2022; Table 7), indicating informality alone does not explain district-level variation after controlling for spatial structure.

Table 7: District-Level Informal Employment and Underemployment Regression

Year	n	Slope	R^2	p
2019	25	0.0279	0.038	0.3508
2022	25	0.0204	0.059	0.2414

5 POLICY RECOMMENDATIONS

Our analysis identifies four high-impact, evidence-based interventions targeting the SHAP-derived drivers of underemployment. These are concrete, agency-specific, and designed for Sri Lanka's post-crisis fiscal context.

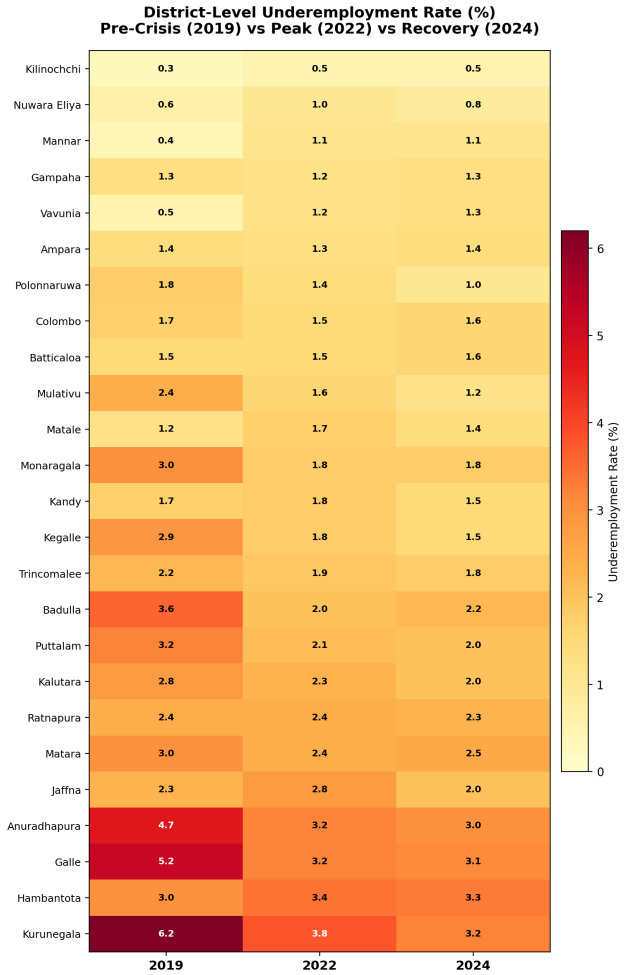

Figure 9: District-level underemployment heatmap (25 districts, 2019/2021/2022). Darker cells indicate higher underemployment. Northern and Southern districts show persistent hotspots.

Formalisation via social protection extension. Informal employment ranks as the second-largest SHAP driver, with a structural break at 2017 indicating formalisation pressures pre-dating the 2022 default. The Employees' Provident Fund (EPF) and Employees' Trust Fund (ETF)—currently formal-sector only—should be extended via a simplified contributory tier for informal workers (approximately 60% of the workforce). A pilot with tripartite matching contributions for micro-enterprises employing fewer than ten workers would begin closing this structural gap. ILO evidence [8] indicates that countries extending social protection to informal workers during crises experienced 15–20% smaller underemployment spikes.

Youth qualification matching. Youth LFPR (SHAP rank 3) shows suggestive Granger causality (p=0.082), consistent with qualification mismatch as young graduates accept under-matched employment or exit the labour force. Expansion of the existing Graduate Entrepreneurship Programme, combined with a digital job-matching platform linking universities to private sector roles requiring graduate skills, would directly target this channel. The World Bank Sri Lanka Human Capital Review [**?**] identifies TVET reform as the highest-priority structural intervention for graduate underemployment reduction.

Remittance stabilisation mechanism. Remittances exhibit a negative lagged association with underemployment; the 2021 collapse from $7.0 bn to $5.5 bn preceded the peak underemployment quarter, consistent with diaspora transfers functioning as a private income buffer. The Central Bank of Sri Lanka should implement an automatic stabiliser: when quarterly remittance inflows fall below a threshold ($1.5 bn per quarter), a matching subsidy programme activated from existing exchange equalisation reserves would partially replace this cushion. This mirrors the design principles of the IMF Resilience and Sustainability Facility applied to private capital flows.

Adoption of ILO labour underutilisation framework. The 2022 LFS artefact—underemployment reported at 2.3% due to social desirability bias during the cash shortage—demonstrates that headline unemployment is an inadequate crisis barometer. The Department of Census and Statistics should adopt the ILO's composite Labour Underutilisation Framework (LU1–LU4 indicators) as quarterly headline statistics alongside unemployment, with explicit disaggregation into time-related and skills-related components. This aligns with the 19th ICLS Resolution [7] and OECD best practice.

Table 8: Policy Mapping to SHAP-Identified Drivers

Recommendation	Target Driver	Agency
EPF/ETF informal extension	Informal emp. (#2)	Labour Ministry
Graduate job platform	Youth LFPR (#3)	Higher Education
Remittance stabiliser	Remittances (lag)	CBSL
ILO LU1–LU4 framework	Measurement artifact	DCS

6 CONCLUSION

Quarterly LFS micro-data reveals crisis underemployment peaks of 32.6%, nine times the concurrent 3.8% unemployment rate, confirming that aggregate unemployment is an inadequate crisis barometer for Sri Lanka. Structural break tests confirm 2021–2022 as a statistically significant rupture in the underemployment–GDP–inflation system, with macro breaks leading the labour market break by one year. The incorporated variables reveal two additional findings: informal employment broke significantly in 2017, predating the crisis and suggesting formalisation pressure as a structural precondition; and part-time employment spiked in 2020 under COVID, acting as a precursor to the 2022 hours-underemployment surge. SHAP (Eq. 5) establishes GDP contraction and informal employment as the dominant drivers, with Youth LFPR, remittances, and part-time employment capturing complementary channels. The interaction model (Eq. 6) confirms structural amplification of GDP elasticity post-2022. The 2–3 quarter TLCC lags are consistent with

a labour hoarding transmission mechanism, and hysteresis analysis indicates that underemployment remained structurally elevated through 2024 despite partial macroeconomic recovery.

ARDL bounds testing on the quarterly dataset (n=36) resolves **RQ4**: the Pesaran F-statistic of 7.25 confirms long-run cointegration at 1%, and the ECM speed-of-adjustment $\hat{\lambda}$=−1.087 (p=0.003) indicates near-complete quarterly correction toward equilibrium. Long-run GDP elasticity is −0.39 pp per 1 pp GDP growth. The sensitivity analysis confirms SHAP rankings are strongly convergent across hours-based and qualification-based definitions ($\rho = 0.893$, $p = 0.007$). Gender-disaggregated analysis reveals distinct driver structures: female underemployment is more inflation-sensitive while male underemployment responds primarily to GDP contractions. The persistent female–male gap ($W = 55.0$, $p = 0.001$) and widening qualification gender gap (0.97 pp to 4.42 pp) indicate the crisis disproportionately affected women's labour market outcomes. District-level analysis confirms substantial spatial heterogeneity that informality alone does not explain. Four targeted policy interventions—EPF/ETF extension, graduate job matching, a remittance stabilisation mechanism, and DCS adoption of the ILO LU1–LU4 framework—address both demand-side and supply-side channels while upgrading statistical infrastructure.

Limitations. This analysis confronts five primary constraints. First, the annual XGBoost model (n=10) imposes a binding degrees-of-freedom constraint that precludes Johansen cointegration testing and full multi-regressor ARDL estimation on the annual dataset; the quarterly panel (n≈43) substantially mitigates this. Second, SHAP values capture predictive association, not causal direction: high informal employment importance may reflect common shocks driving both informality expansion and hours contraction simultaneously rather than a direct causal path. Third, real wages and sectoral employment shares (manufacturing, services, agriculture) were excluded due to insufficient quarterly availability for 2015–2025, despite being theoretically important transmission channels. Fourth, the 2022 LFS measurement artefact compromises annual inference for the hours-based series even with the quarterly workaround; results for that year should be interpreted with caution. Fifth, Sri Lanka represents a specific small open economy configuration (sovereign default combined with high remittance dependence), which may limit external validity to other developing countries absent a comparable crisis-remittance constellation. The ECM speed-of-adjustment $|\hat{\lambda}|>1$ indicates mild overshooting, consistent with volatile crisis-period dynamics, and should be interpreted cautiously.

Future work will: (i) re-estimate ARDL/ECM on the expanded quarterly dataset (n≈43) once 2025 Q3–Q4 LFS data are released; (ii) incorporate the 2025 annual LFS data; (iii) extend gender analysis to quarterly disaggregation for higher-power break detection; and (iv) develop district-level panel models to explain spatial heterogeneity beyond informality.

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